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IoT-Based Intelligent Crop Monitoring System

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PHASE I: PROJECT OVERVIEW

- **Phase I – Research & System Design**
- Phase I focused on establishing the foundation of the IoT-Based Intelligent Crop Monitoring System.
- **Major Activities:**
- Smart Agriculture study
- Existing technology analysis
- Problem identification
- Literature review
- System objectives
- IoT architecture design
- Hardware and software selection
- Feasibility analysis
- Prototype planning
- **Goal:** Develop an affordable, scalable and intelligent farming solution.

SMART AGRICULTURE

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What is Smart Agriculture?

Smart Agriculture uses modern technologies such as IoT, AI, ML, sensors, cloud computing and data analytics to improve farming.

It enables:

- Real-time field monitoring
- Precision irrigation
- Targeted fertilizer application
- Early problem detection
- Better crop management
- Reduced resource wastage
- Improved productivity

The project aims to shift farming from traditional intuition-based decisions toward **data-driven precision agriculture**.

MOTIVATION & GOAL

Agriculture faces increasing challenges from:

- Population growth
- Climate change
- Water scarcity
- Soil degradation
- Limited agricultural land
- Increasing food demand

Advanced farming technologies can help, but many solutions remain expensive or complex.

Project Goal:

Develop an affordable intelligent system that improves crop productivity while reducing resource consumption and environmental impact.

Core Principle:

Produce More with Less.

EXISTING TECHNOLOGIES

Modern precision agriculture uses:

IoT Sensors

- Soil moisture
- NPK
- pH
- Temperature
- Humidity
- Rainfall

AI & ML

- Data analysis
- Prediction
- Risk detection
- Recommendations

Farm Management Platforms

- Centralized data
- Monitoring dashboards
- Decision support

However, farmers may still receive large amounts of raw data without clear actions.

IDENTIFIED GAPS

Major Gaps in Existing Systems

1. Expensive & Complex

Many solutions are difficult for small and medium farms to adopt.

2. Data Fragmentation

Sensor, weather and other data sources may remain separated.

3. Actionability Gap

Farmers receive numbers and charts but not always clear recommendations.

Proposed Approach

Convert:

Raw Data → Analysis → Actionable Recommendation

This makes technology more useful for practical farm management.

PROBLEM STATEMENT

The project addresses four major agricultural problems:

1. Inefficient Resource Management

Excessive water, fertilizer and pesticide usage.

2. Delayed Problem Detection

Problems may be identified only after visible crop damage.

3. Lack of Hyper-Local Data

General information cannot represent individual field conditions.

4. Yield & Income Uncertainty

Poor management can reduce productivity and financial stability.

Core Problem:

Farmers need timely, localized and actionable information.

PROJECT OBJECTIVES

Main Objectives

Continuously monitor soil and environmental conditions
Collect real-time field data
Process sensor information
Generate automated alerts
Provide predictive insights
Support precision irrigation
Support targeted fertilizer usage
Provide actionable recommendations
Improve resource efficiency
Promote sustainable farming

Parameters Monitored

Soil Moisture | pH | NPK | Temperature | Humidity | Rainfall

PROPOSED SYSTEM ARCHITECTURE

Four-Layer Architecture

Layer 1 – Sensing

Sensors collect field data.

↓

Layer 2 – Edge Processing

ESP32 and Raspberry Pi process and transmit data.

↓

Layer 3 – Cloud & Intelligence

Data is stored and analyzed.

↓

Layer 4 – Application

Farmers receive dashboards, alerts and recommendations.

Overall Flow

Sensors → ESP32 → Raspberry Pi → Database/Cloud → AI → Mobile App → Farmer

HARDWARE & SOFTWARE

Hardware

- Soil Moisture Sensor
- pH Sensor
- NPK Sensor
- DHT22 Sensor
- Rain Detection Sensor
- ESP32
- Raspberry Pi 4

Software

- Python
- AI/ML
- Mobile Application
- Cloud Platform
- Database

Technologies

ESP32: Data acquisition

Raspberry Pi: Processing and gateway

Python: Backend and AI

Mobile App: Farmer interaction

SYSTEM DATA FLOW

Data Flow
Field Sensors
↓
ESP32
Data acquisition and preprocessing
↓
Raspberry Pi 4
Data aggregation
↓
Database / Cloud
Data storage
↓
Python Backend / AI
Analysis and intelligence
↓
Mobile Application
Dashboard, alerts and recommendations
↓
Farmer
The system transforms physical field conditions into useful decision-making information.

FEASIBILITY ANALYSIS

Technical Feasibility
The system uses commercially available IoT components such as ESP32, Raspberry Pi and sensors.
Economic Feasibility
The architecture is designed to remain affordable for small and medium-sized farms.
Operational Feasibility
The application simplifies complex agricultural data into understandable information.
Expected Benefits

- Reduced water wastage
- Reduced fertilizer wastage
- Improved monitoring
- Faster intervention
- Better productivity
- Sustainable farming

Phase I established the proposed system as technically and economically viable.

PHASE I CONCLUSION

Phase I Achievements

✓ Literature and technology reviewed
✓ Agricultural problems identified
✓ Project objectives defined
✓ System architecture designed
✓ Hardware and software selected
✓ Feasibility evaluated
✓ Prototype approach planned

Transition to Phase II

Phase I answered:

What problem are we solving?
How should the system be designed?

Phase II focuses on building and validating the proposed solution.

PHASE II: IMPLEMENTATION

Phase II Focus

Phase II transformed the proposed design into a working prototype and software interface.

Major Activities

Physical prototype development
Sensor integration
ESP32 implementation
Environmental data collection
CROP SAMS application
Dashboard development
Alerts and recommendations
Data-flow implementation
Prototype validation

Phase II Goal:

Demonstrate that the proposed IoT architecture can operate as a practical intelligent crop monitoring system.

PHYSICAL PROTOTYPE

Prototype Components

The prototype uses an **ESP32 microcontroller** connected to commercial agricultural sensors.

Sensors:

Capacitive Soil Moisture
DHT22 Temperature & Humidity
Rain Detection
NPK
pH

Working Principle

Sense → Collect → Process → Analyze → Alert

The prototype was tested in an outdoor garden environment to evaluate practical sensor data collection.

CROP SAMS APPLICATION

CROP SAMS

The CROP SAMS mobile application provides the user-facing interface of the system.

Main Purpose

Transform raw agricultural data into **understandable and actionable information**.

Main Features

Environmental monitoring
Dashboard
Alerts
Crop information
Recommendations
AI assistance
Multilingual support

The application forms the connection between the intelligent system and the farmer.

MULTILINGUAL SUPPORT

Supported Languages
English
Hindi – हिन्दी
Gujarati – ગુજરાતી
Marathi – मराठी
Why?
Language barriers can prevent farmers from effectively using digital agricultural systems.
Benefit
Multilingual interaction improves:
Accessibility
Understanding
Usability
Farmer adoption
The application allows farmers to interact with the system using their preferred supported language.

GEO-LOCATION & FARM INFORMATION

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Farmer & Farm Profile

The application collects information related to:

Geo-location

Land

Soil

Available resources

Farm conditions

Purpose

This information helps the system provide recommendations based on the farmer's specific agricultural context.

Key Concept

Hyper-Local Information → Better Decision Support

The Phase-II application includes dedicated interfaces for geo-location, land and soil information, and resource selection.

CROP RECOMMENDATION

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Crop Recommendation System

The system combines farmer and farm information to support crop selection.

Inputs

Farm profile
Land information
Soil conditions
Environmental data
Historical information
Available resources

Output

Personalized Crop Recommendation

The system integrates a Google Generative AI service into the recommendation workflow to support personalized agricultural guidance.

AI ASSISTANT

AI-Based Agricultural Assistance
The AI Assistant provides an intelligent interaction layer between the farmer and the system.
Process
Farmer / Sensor Data
↓
Historical Data
↓
AI Recommendation Engine
↓
Contextual Analysis
↓
Recommendation
Purpose
Decision support
Agricultural guidance
Personalized recommendations
Historical-data-based insights
The system uses farmer profiles and historical trends to support recommendations.

ALERT & NOTIFICATION SYSTEM

Intelligent Alerts
The system analyzes sensor and external conditions to identify important events.
Alert Process
Sensor Data
→ Threshold Analysis
→ Weather Monitoring
→ Alert Decision
→ Farmer Notification
Example Alerts
Critical soil moisture
High humidity risk
Critical NPK
Rain detection
Irrigation requirement
The application focuses on high-priority events requiring farmer attention.

ENVIRONMENTAL DASHBOARD

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Dashboard Monitoring

The dashboard converts sensor readings into actionable information.

Temperature: 30.2°C

→ Heat and disease risk analysis

Humidity: 21.2%

→ Pest and fungal monitoring

Soil Moisture: 0%

→ Critical irrigation alert

Rain Status: No Rain

→ Irrigation decision support

Charts: Historical trends and anomalies

This demonstrates the system's ability to transform raw readings into understandable guidance.

DFD & SYSTEM VALIDATION

Implemented Data Flow
Sensor / Farmer Input
↓
Data Processing
↓
Historical Database
↓
AI Recommendation
↓
Alert & Scheduling
↓
Reports & Analytics
↓
CROP SAMS
Validation
The implementation demonstrates:
Sensor data processing
Historical storage
Alert generation
Crop management
Recommendations
Report generation
The Level-2 DFD defines these functional processes.

RESULTS & QUANTIFIED BENEFITS

Targeted Project Outcomes
15–20%
Increase in crop yield and quality
Up to 30%
Reduction in water consumption
Reduced Fertilizer Waste
Through targeted nutrient application
Proactive Management
Real-time insights and automated alerts
Overall Impact
The system aims to improve:
Productivity
Profitability
Resource efficiency
Sustainability
Decision-making
These values are project targets/expected benefits defined in the report.

CONCLUSION & FUTURE SCOPE

Conclusion

The project demonstrates how **IoT + AI + Mobile Technology** can transform agricultural monitoring.

Raw Data → Intelligence → Action

Future Scope

Automatic water pumps
Solenoid valves
Automatic fertilizer dispensing
Camera-based disease detection
Improved AI models
Farmer feedback integration
LoRaWAN / NB-IoT
Solar-powered field nodes

Final Vision

From Reactive Farming to Proactive, Data-Driven Agriculture.

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THANK YOU